C Programming Practice Worksheet: If-Else and For Loops

For Grade 7 Students

Practice for your C programming exam! This worksheet helps you practice `if-else` statements and `for` loops. Write your answers neatly in the spaces provided. Good luck!

Instructions

- **Part A**: Write complete C programs for each question. Include `#include <stdio.h>` and the `main` function. Test your logic by thinking through what the program does.
- **Part B**: For each buggy code, read what it's supposed to do, find the errors, list them, and write the corrected code.
- **Tips**: Check for missing semicolons, braces, or `&` in `scanf`. Make sure loops and conditions are correct. Write clearly!
- **Example Program Structure**:

```
#include <stdio.h>
int main() {
    // Your code here
    return 0;
}
```

Part A: Coding Problems

Solve each problem by writing a complete C program. Use the space below each question to write your code.

1. Positive or Negative

Write a program to input a number and print "Positive" if the number is greater than 0, otherwise print "Negative or Zero".

2. Print Numbers
Write a program to print all numbers from 1 to 10 using a `for` loop.

3. Even or Odd
Write a program to input a number and print "Even" if it's divisible by 2, else print "Odd".

4. Sum of Numbers
Write a program to calculate and print the sum of numbers from 1 to 5 using a `for` loop.

5. Grade Checker
Write a program to input a student's score (0–100) and print "Pass" if the score is 60 or above, else print "Fail".

6. Print Multiples
Write a program to print all multiples of 3 between 1 and 15 using a `for` loop.

7. Count Even Numbers
Write a program to count how many even numbers are between 1 and 10 using a `for` loop and an `if` statement, then print the count.

8. Average and Pass/Fail
Write a program to input 3 test scores, calculate their average using a `for` loop, and print "Good" if the average is 80 or above, else print "Need Improvement".

9. Largest of Two
Write a program to input two numbers and print the larger one using `if-else`.

10. Sum of Odd Numbers
Write a program to calculate and print the sum of all odd numbers between 1 and 20 using a `for` loop and an `if` statement.

Part B: Debugging Exercises

Each program below has 1–2 errors. Read what the program is supposed to do, find the errors, list them, and write the corrected code in the space provided.

1. Print "Adult" or "Child"
Goal: Input an age and print "Adult" if age is 18 or above, else print "Child".
Buggy Code:

```
#include <stdio.h>
int main() {
    int age
    printf("Enter age: ")
    scanf("%d", age);
    if (age >= 18) {
        printf("Adult\n");
    } else {
        printf("Child\n");
    }
    return 0;
}
```

Errors Found:

Corrected Code:

2. Print Numbers 1 to 5
Goal: Print numbers from 1 to 5.
Buggy Code:

```
#include <stdio.h>
int main() {
    for (int i = 1; i <= 5; i++) {
        printf("%d\n", i);
    return 0;
}
```

Errors Found:

Corrected Code:

3. Check if Number is Even
Goal: Input a number and print "Even" if it's even, else print "Odd".
Buggy Code:

```
#include <stdio.h>
int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    if (num % 2 = 0) {
        printf("Even\n");
    } else {
        printf("Odd\n");
    }
    return 0;
}
```

Errors Found:

Corrected Code:

4. Sum of 1 to 10
Goal: Calculate and print the sum of numbers from 1 to 10.
Buggy Code:

```
#include <stdio.h>
int main() {
    int sum;
    for (int i = 1; i <= 10; i++) {
        sum = sum + i;
    }
    printf("Sum is %d\n", sum);
    return 0;
}
```

Errors Found:

Corrected Code:

5. Print Pass/Fail for 3 Scores
Goal: Input 3 scores and print "Pass" if each score is 50 or above, else print "Fail".
Buggy Code:

```
#include <stdio.h>
int main() {
    int score;
    for (int i = 1; i <= 3; i++) {
        printf("Enter score %d: ", i);
        scanf("%d", &score);
        if (score >= 50)
            printf("Pass\n");
            printf("Fail\n");
    }
    return 0;
}
```

Errors Found:

Corrected Code: